

Abstract: RoboHitch enables a single-arm robot to tie hitch knots without ordered topology tracking. It learns from human demonstrations using only disordered 3D rope keypoints and RGB images. Dynamic GAE, CAE, and bidirectional cross-attention jointly predict pick, rotation, and place affordances for real-world single-arm knot tying.

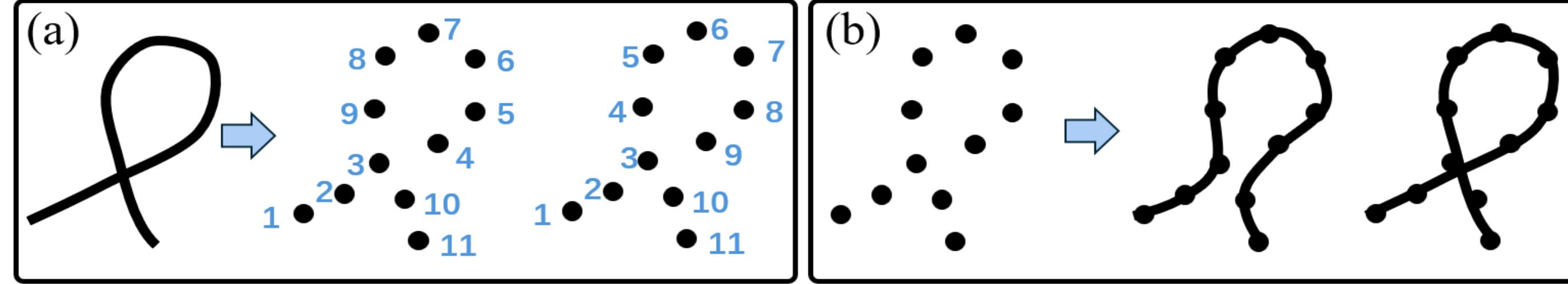
Motivation

How to represent rope state during tying?

RGB/PointCloud	Easily Estimation	Less Structural
Keypoint	Hard to Track	Structural

Core challenge: Knot-tying tasks induce frequent rope bending and crossings. Most existing methods rely on **tracking ordered keypoints and explicit edge connectivity** to represent the rope state, which is **fragile** under deformation, leading to erroneous state estimation and downstream manipulation failures.

Order-agnostic rope keypoints representation without tracking?



Problem 1: Point Set Permutation

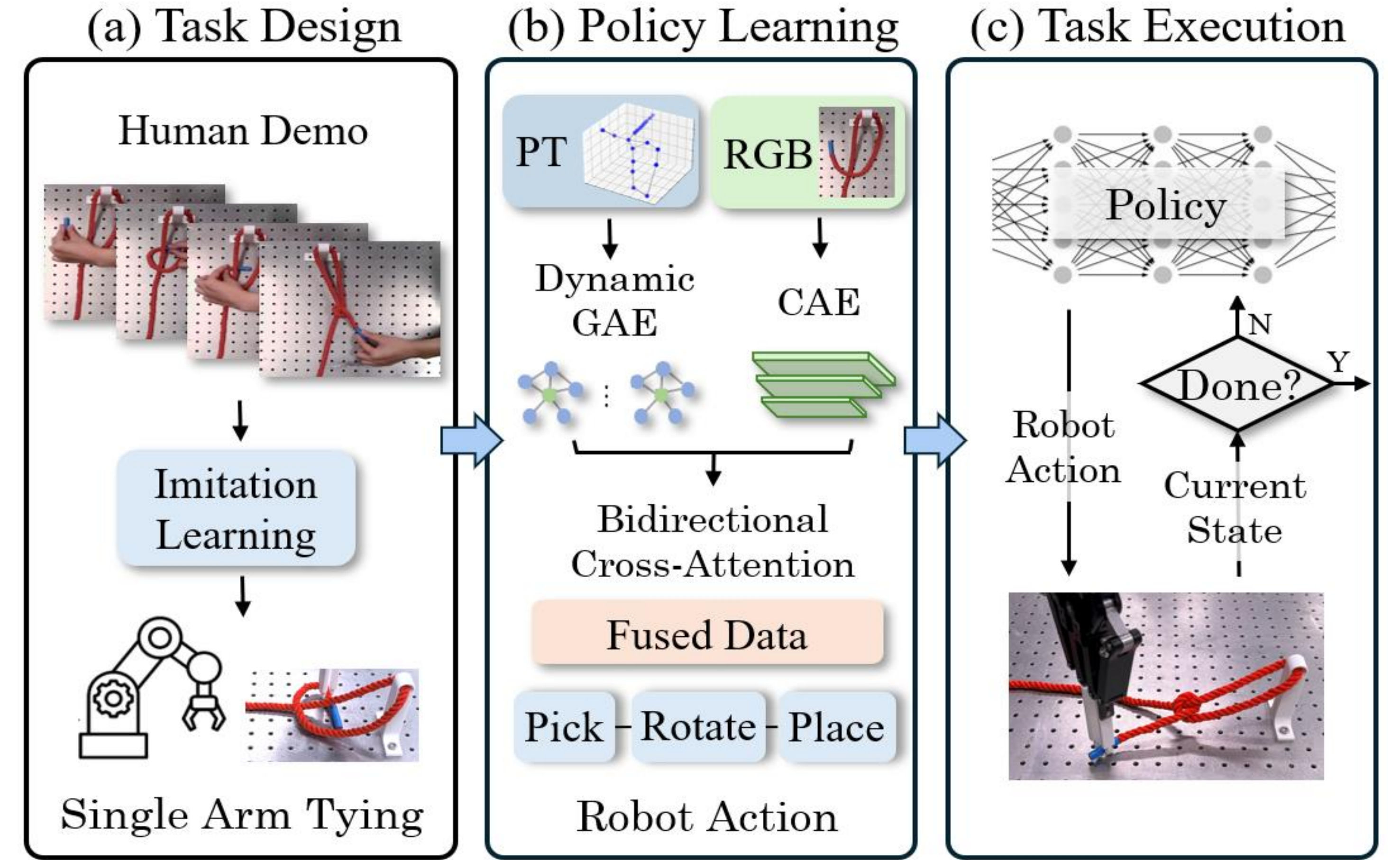
Identical states can result in different keypoint sequences.

Problem 2: Topological Ambiguity

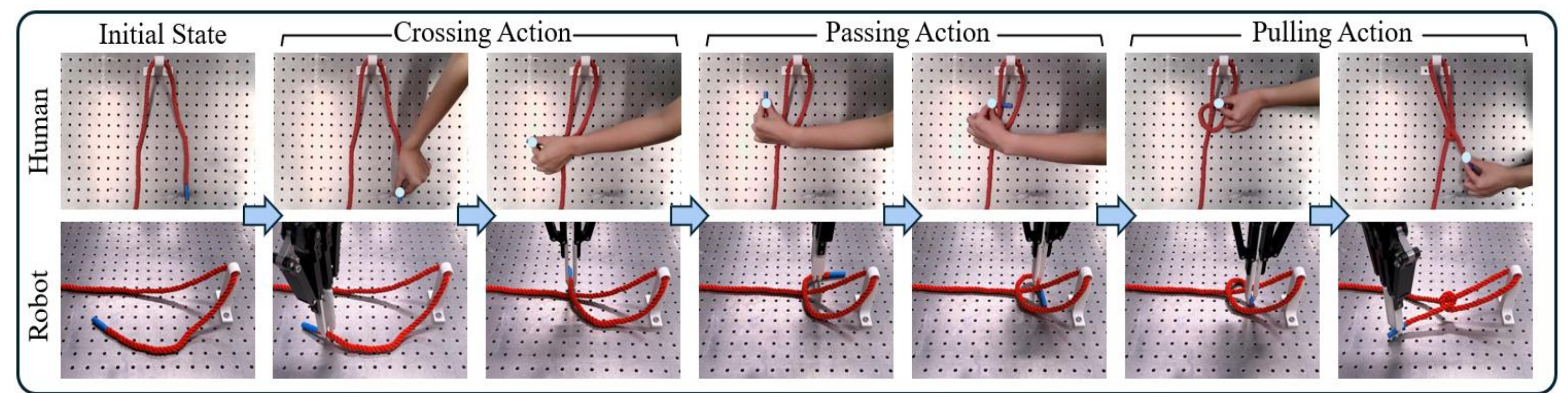
Multiple rope configurations may correspond to the same keypoint set.

Insight:

Disordered keypoints (geometry feature) + RGB image (scene feature) help introduce stronger priors for more efficient learning without precise tracking.



Learning from Demonstrations



Human hand motions are converted into expert trajectories; robot predicts actions online from visual observations.

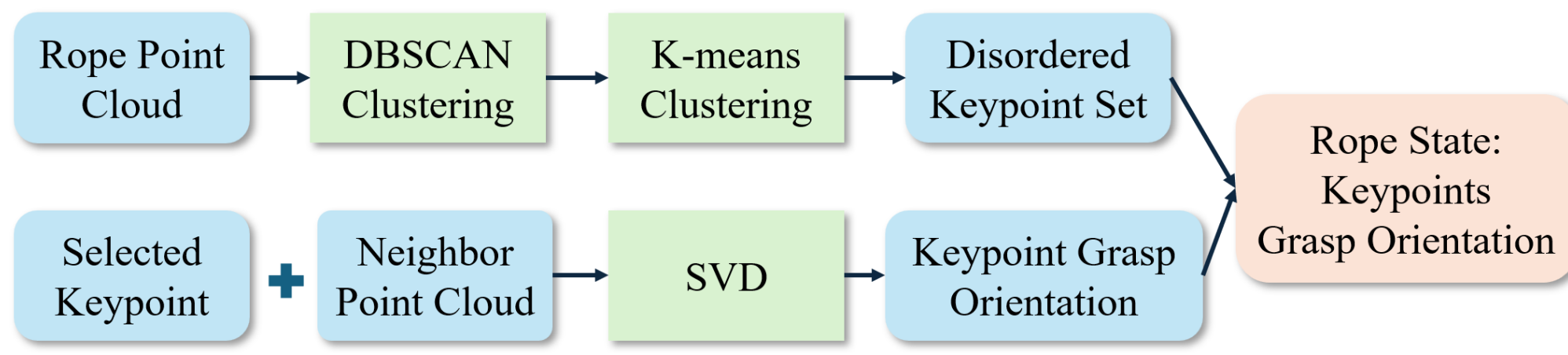
We collect human demonstration videos and use MediaPipe to label pick points, in-hand rotations, and place positions of robot action.

- Pick Label:** Nearest rope keypoint
- Rotation Label:** Finger orientation
- Place Label:** Gaussian heatmap over image

$$\mathcal{L} = \lambda_{\text{pick}} \mathcal{L}_{\text{pick}} + \lambda_{\text{rot}} \mathcal{L}_{\text{rot}} + \lambda_{\text{place}} \mathcal{L}_{\text{place}}$$

RoboHitch Framework

Clustering-based Rope State Estimation



RoboHitch represents the rope as unordered 3D keypoints, instead of a tracked topological chain, enabling robust perception under rope deformation and self-occlusion.

Multimodal Feature Encoding

Graph Autoencoder (GAE):

To address the **Point Set Permutation** problem, we employ a dynamic permutation-invariant GAE. It learns consistent latent encodings regardless of keypoint order by employing symmetric message passing with shared MLP parameters and aggregation functions:

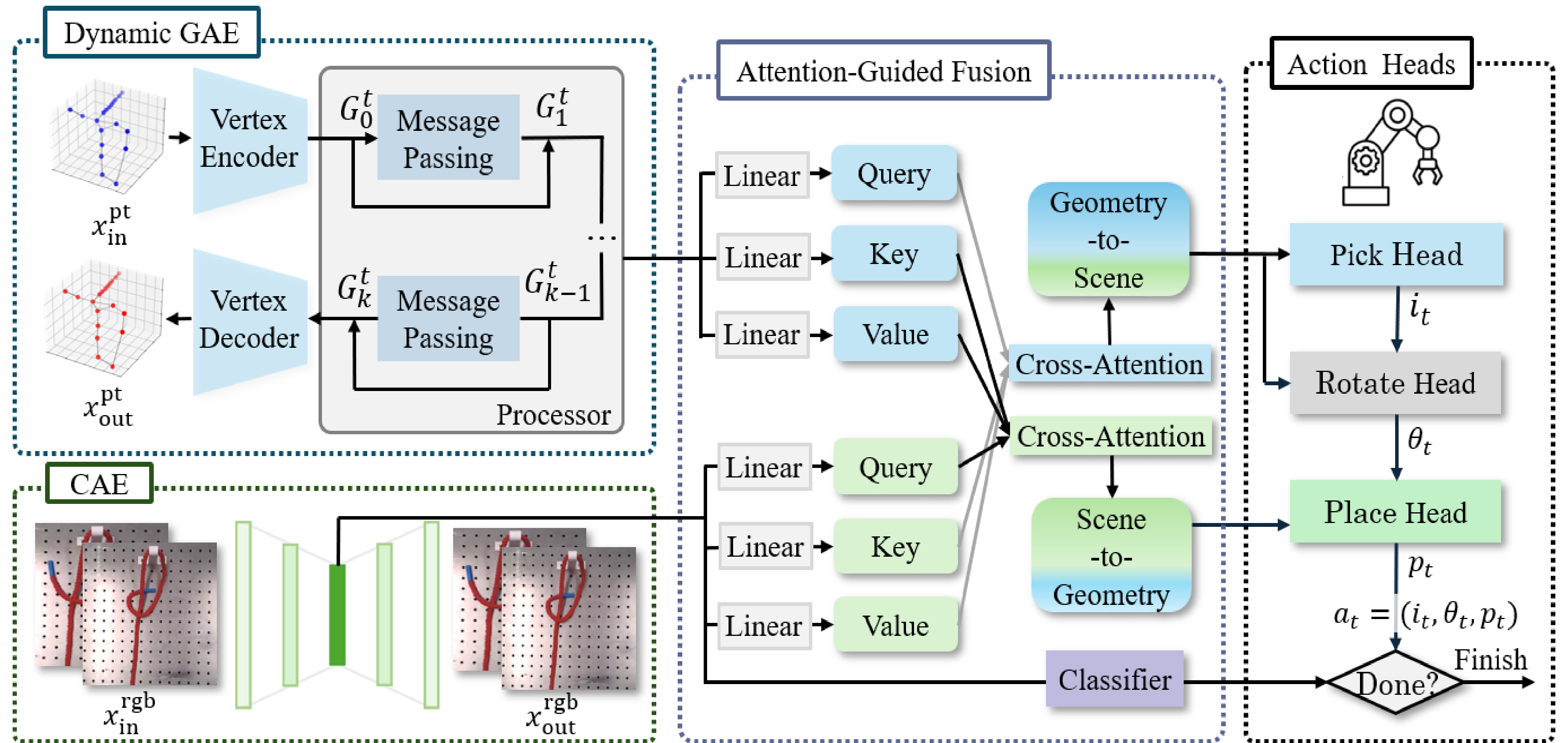
$$x'_i = \text{AGG}_{j \in \mathcal{N}(i)} (\text{MLP}(x_i \parallel (x_j - x_i)))$$

The GAE is trained via reconstruction loss to capture rope topology.

Image Autoencoder (CAE):

we utilize a CAE to encode raw visual observations into a compact latent space, optimized via reconstruction loss.

$$\phi^{I*}, \psi^{I*} = \arg \min_{\phi^I, \psi^I} \|I_t - (\phi^I \circ \psi^I)(I_t)\|^2$$



Bidirectional Cross-Attention Modal Fusion

To address the **Topological Ambiguity** problem, we employ a **bidirectional cross-attention** mechanism to correlate keypoints (geometry feature) with image (scene feature):

Geometry $\rightarrow$ Scene: Solve the problem of *where to pick*.
Scene $\rightarrow$ Geometry: Solve the problem of *where to place*.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Action Affordance Prediction

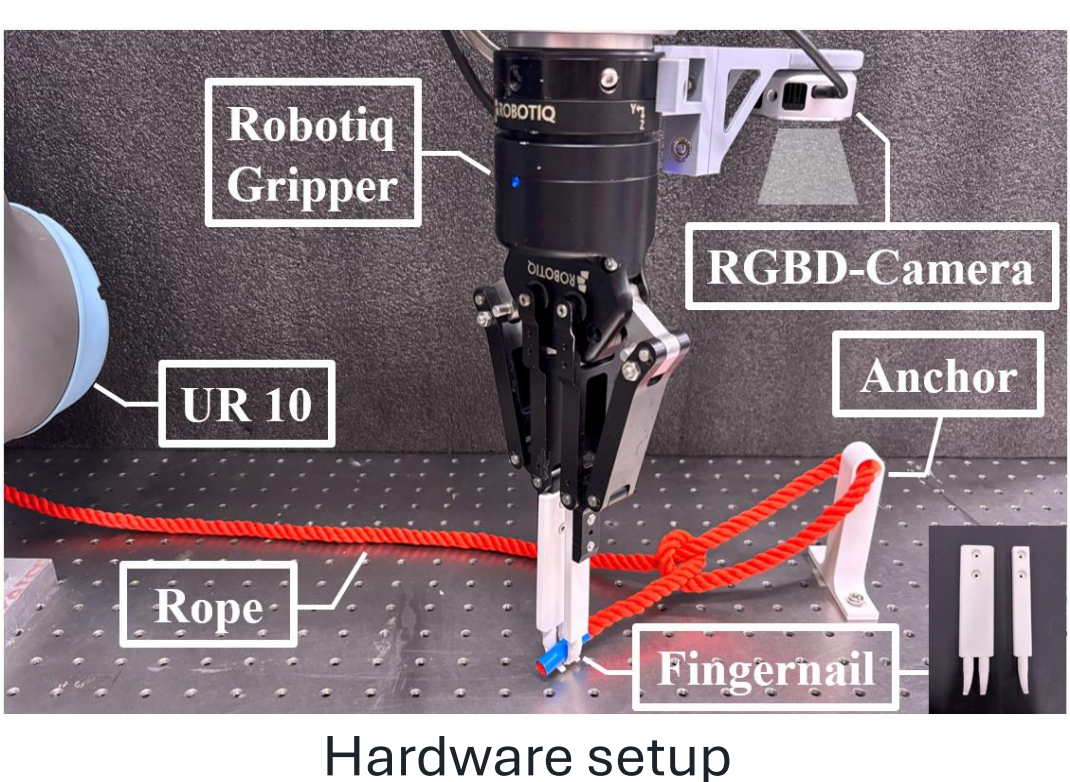
We decompose the joint distribution to model actions without exponential dimensionality growth:

$$a_t = (i_t, \theta_t, p_t), \quad P(i_t, \theta_t, p_t) = P(i_t)P(\theta_t | i_t)P(p_t | i_t, \theta_t)$$

i_t : pick keypoint θ_t : in-hand rotation p_t : place pixel

An MLP-based classifier is used to infer task termination

Experimental Results

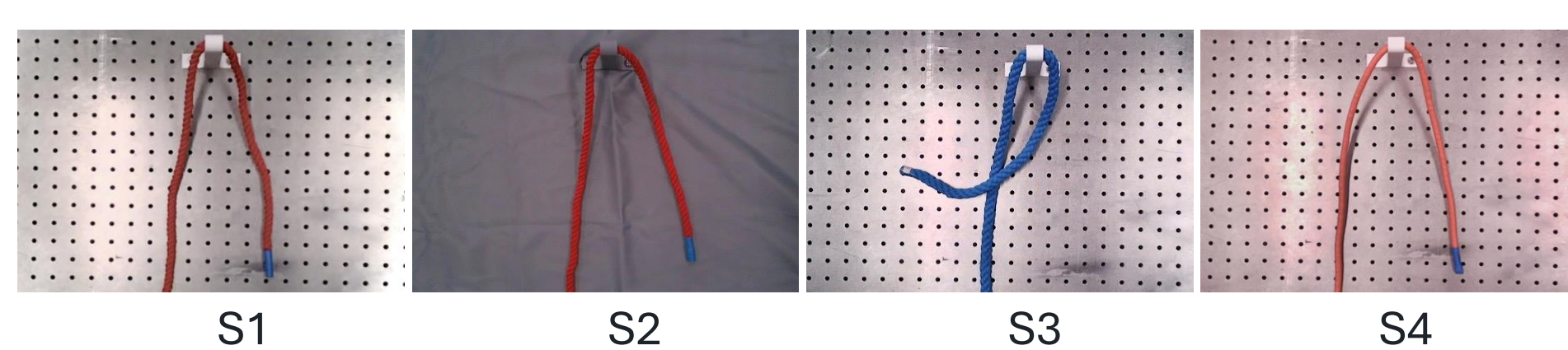


84%
Success Rate
42 / 50 trials

Reference	SR
Dinkel <i>et al.</i>	50%
Nair <i>et al.</i>	38%
Priya <i>et al.</i>	66%
Pathak <i>et al.</i>	60%
RoboHitch(Ours)	84%

- 6-DOF UR10 manipulator;
- Robotiq 2F-140 gripper;
- Wrist-mounted Realsense L515 RGB-D camera;
- Bio-inspired fingernails for reliable rope grasping.

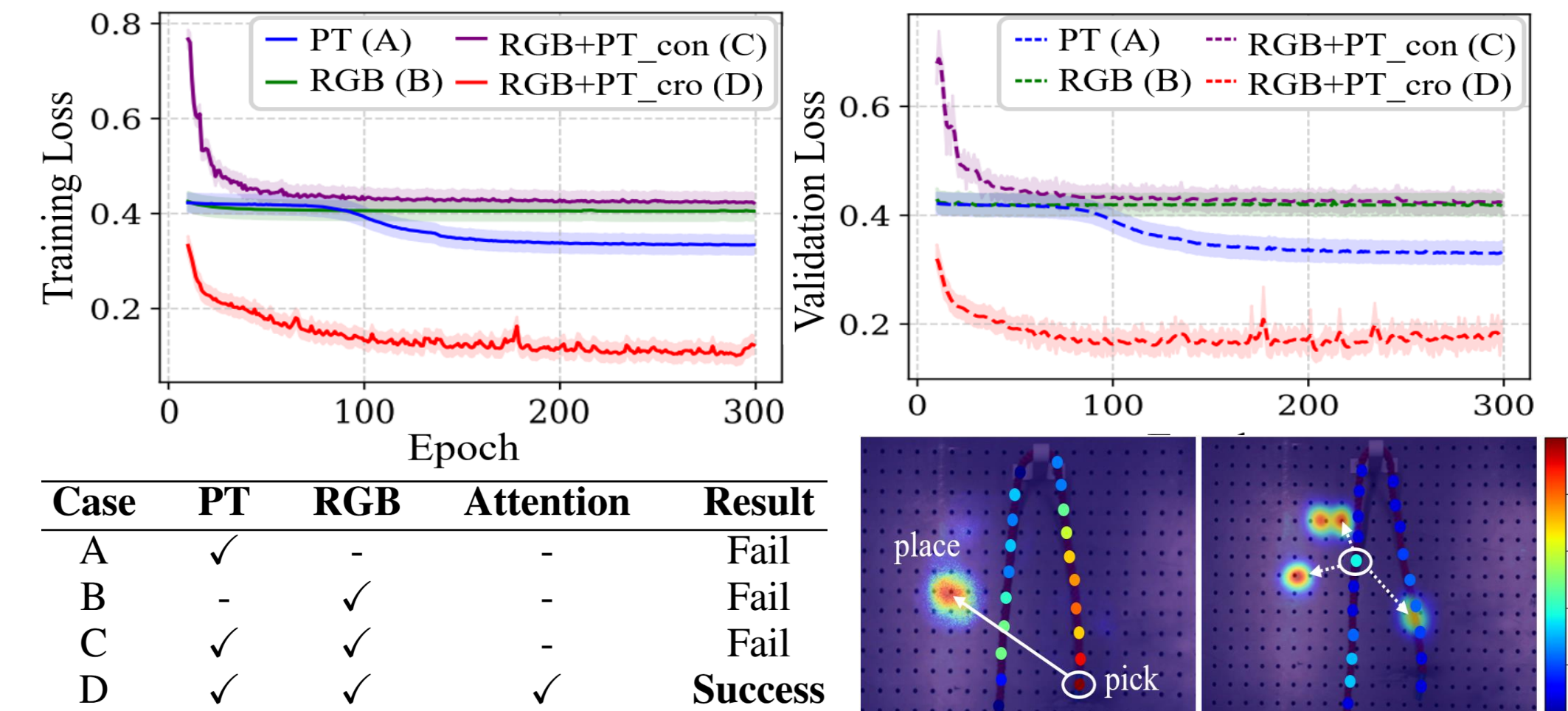
Generalization



Scenario	Diameter	Material	Background	Success
S1	10mm	Nylon	seen	5/5
S2	10mm	Nylon	unseen	4/5
S3	14mm	Nylon	seen	3/5
S4	10mm	Polypropylene	seen	0/5

Stiffer ropes reduce performance due to elastic restoration.

Ablation Study



Keypoints localize grasps, RGB provide scene context, and cross-attention aligns both modalities for robust affordance prediction.